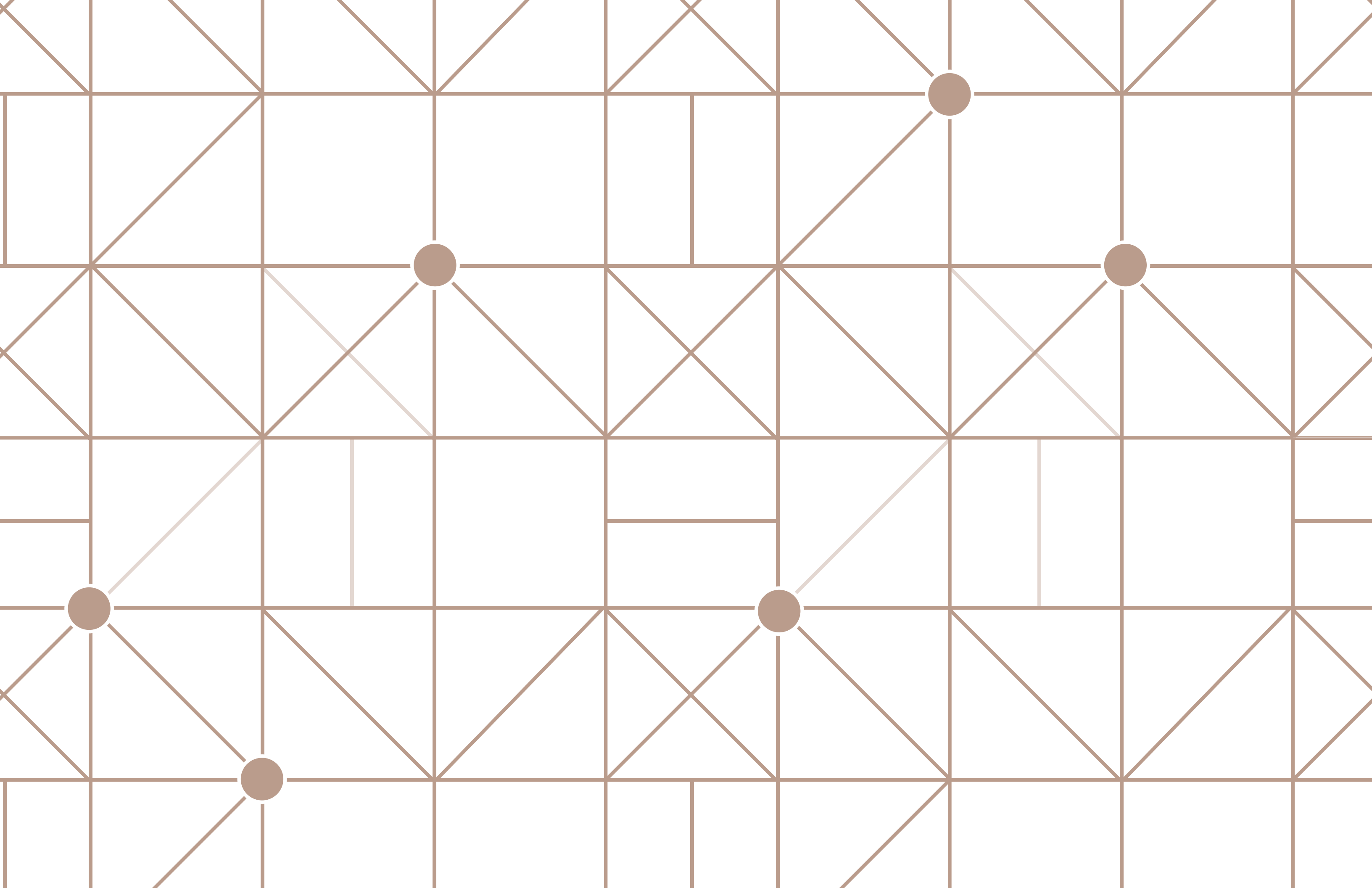
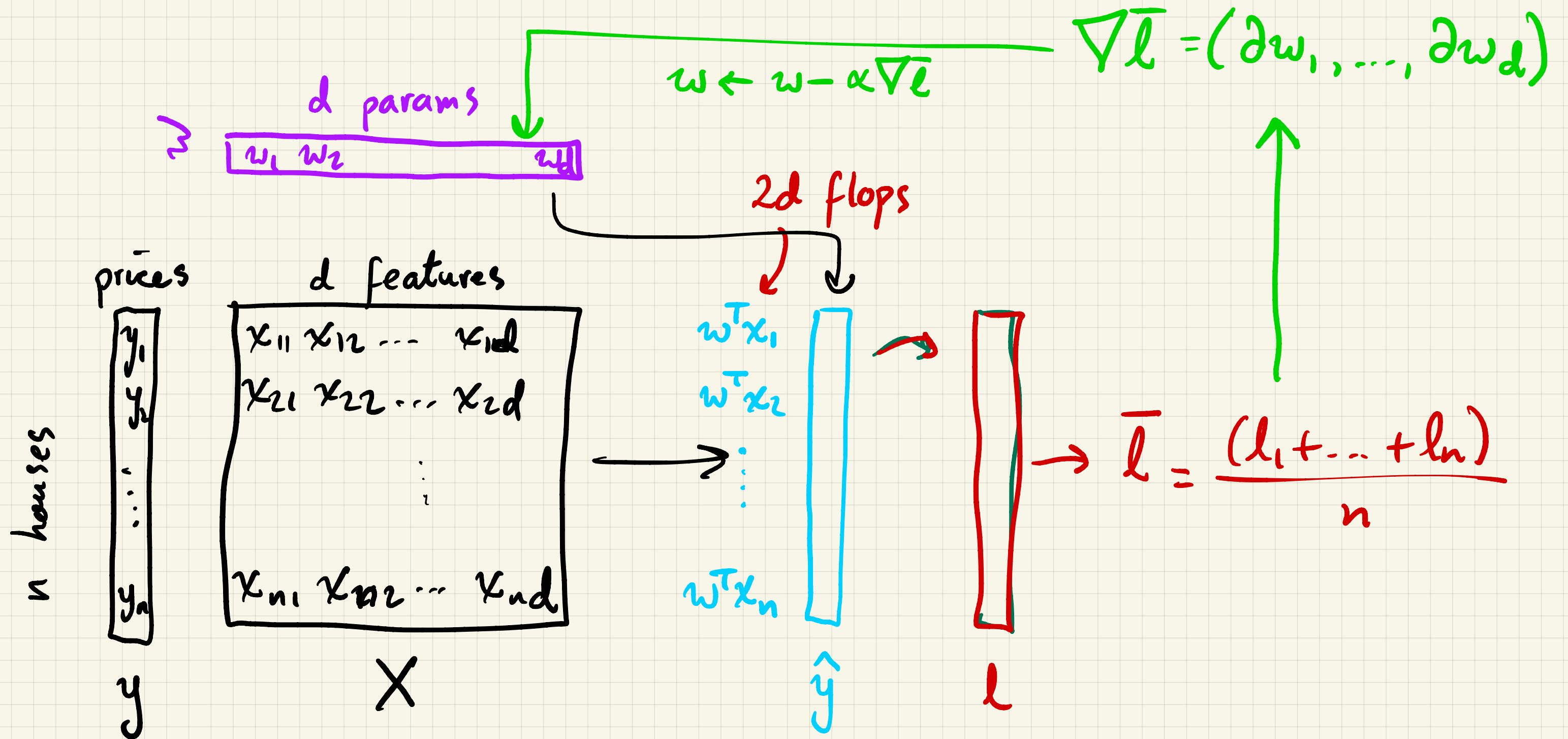
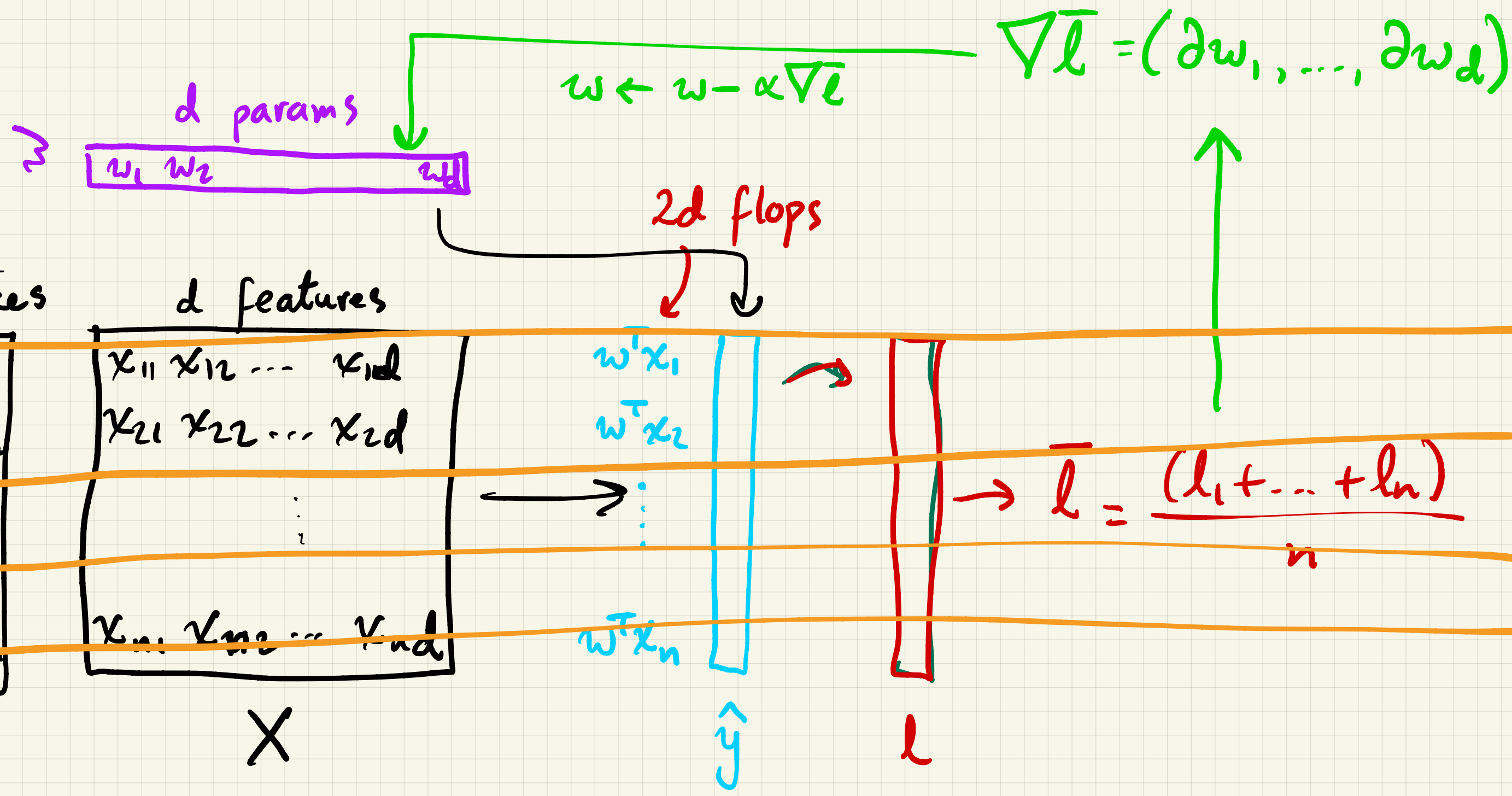




Learning house prices using lin reg & grad descent

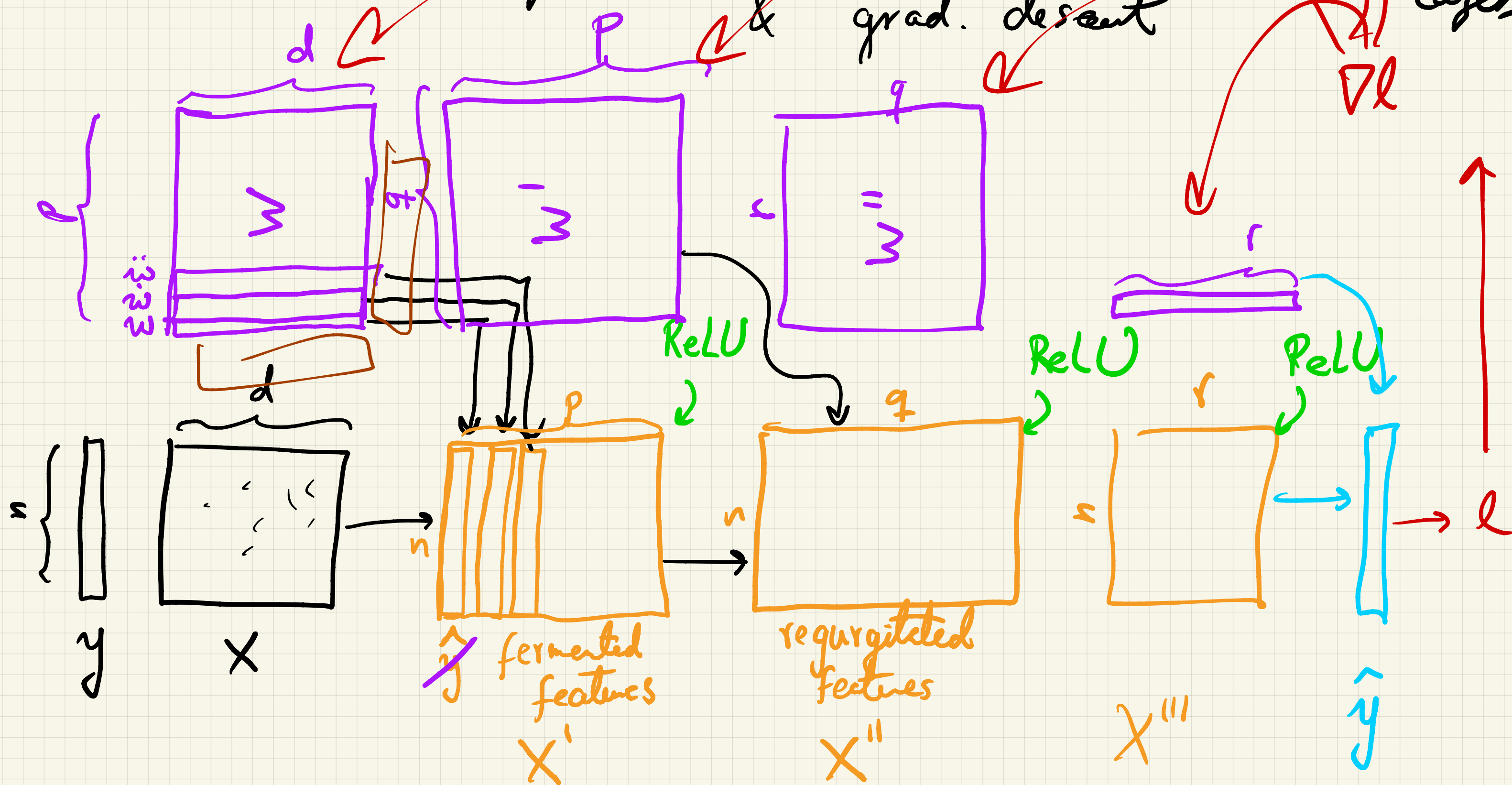


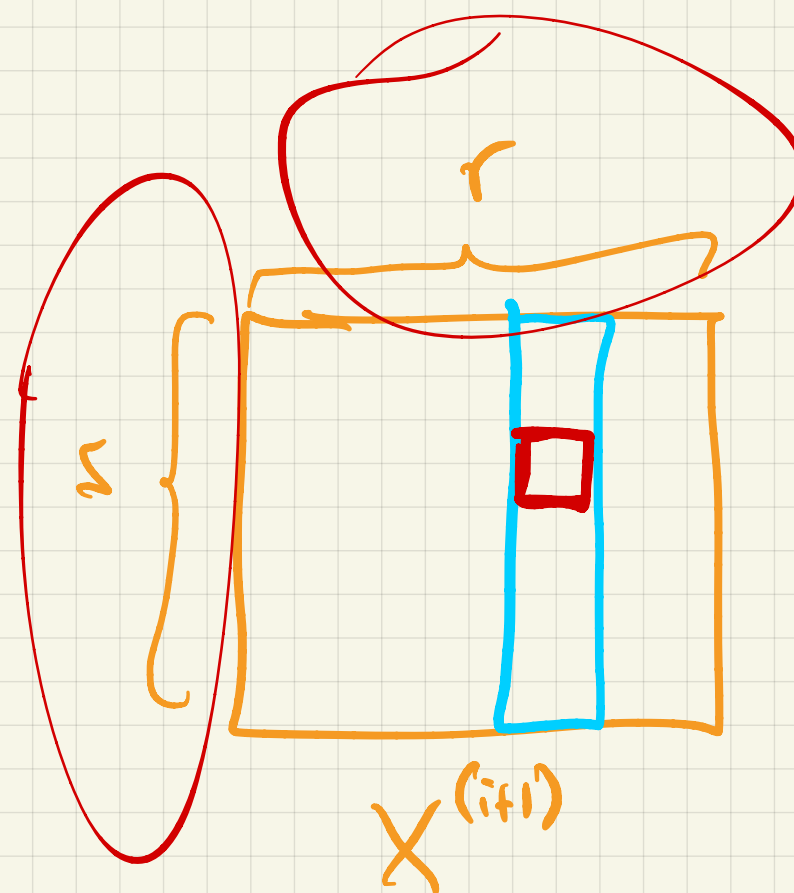
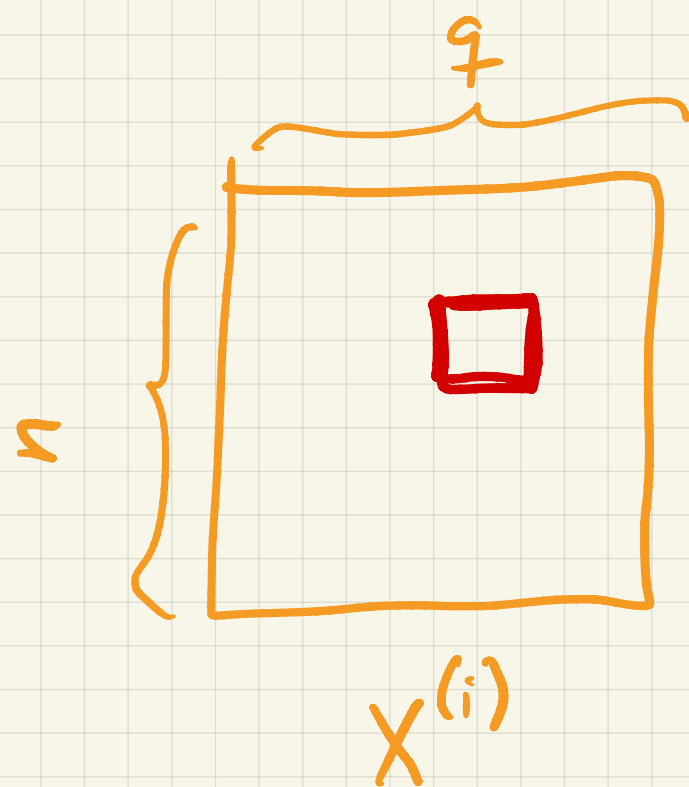
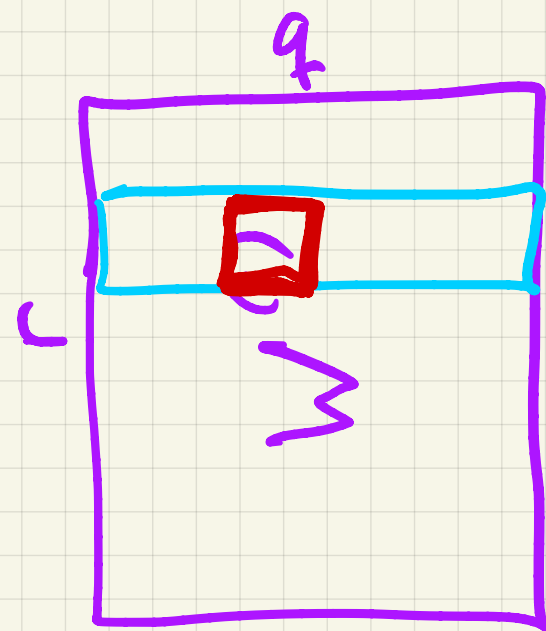
Learning house prices using lin reg & grad descent



Part II

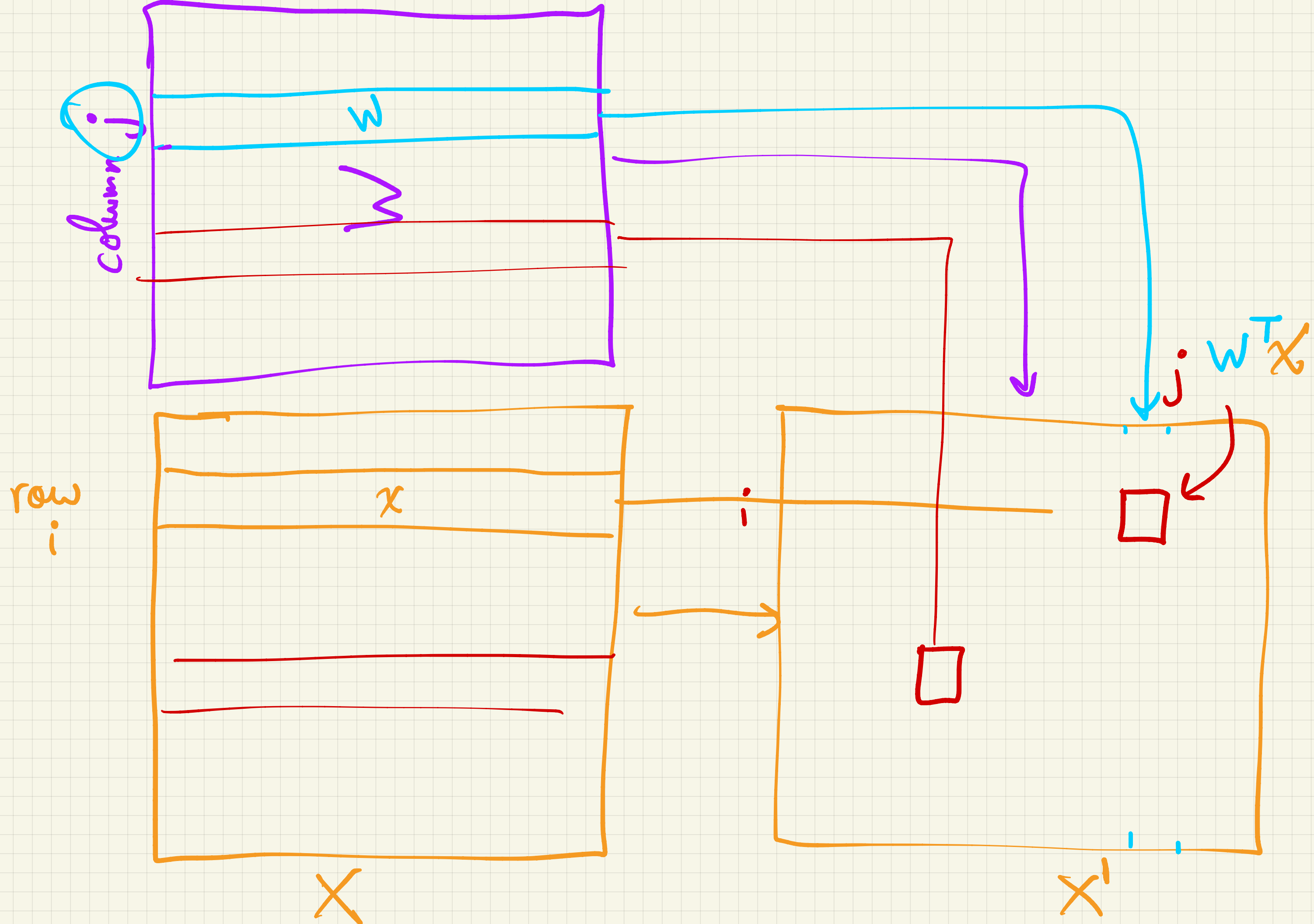
Learning house prices using NNs w/ 4 hidden layers & grad. descent

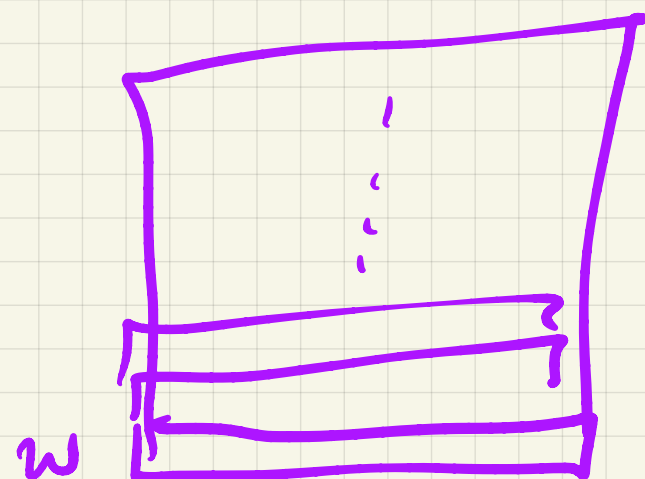




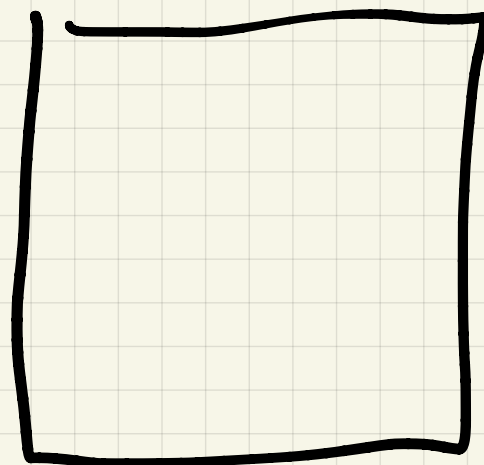
Max parallel
 $n \times r$ GPUs

each GPU needs
 only one
 row of X &
 one col of

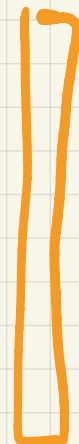




y



x



y

Part II

Learning LLMs using
Transformers & grad. descent

⚠ $w \neq d$

tokens in vocabulary

w

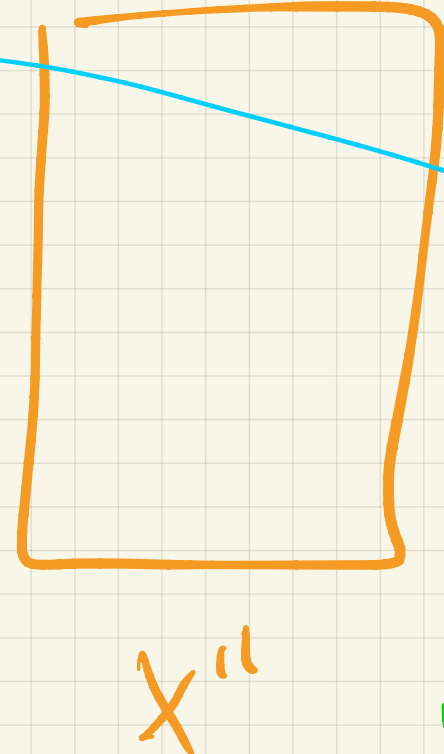
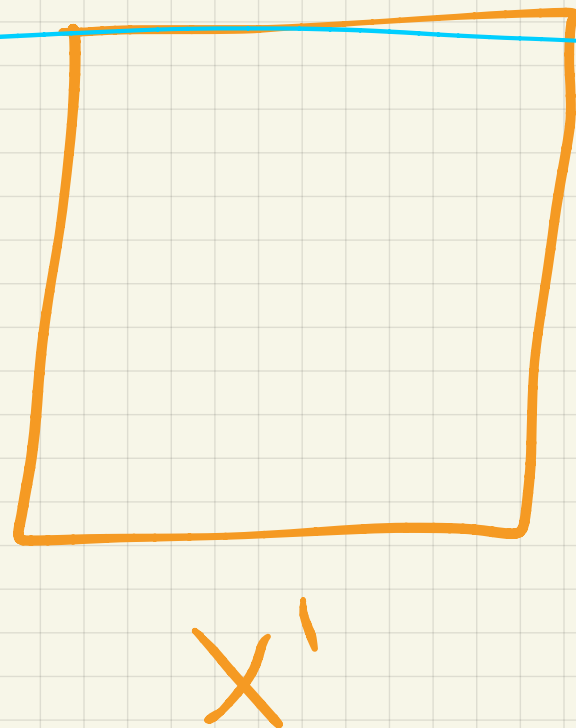
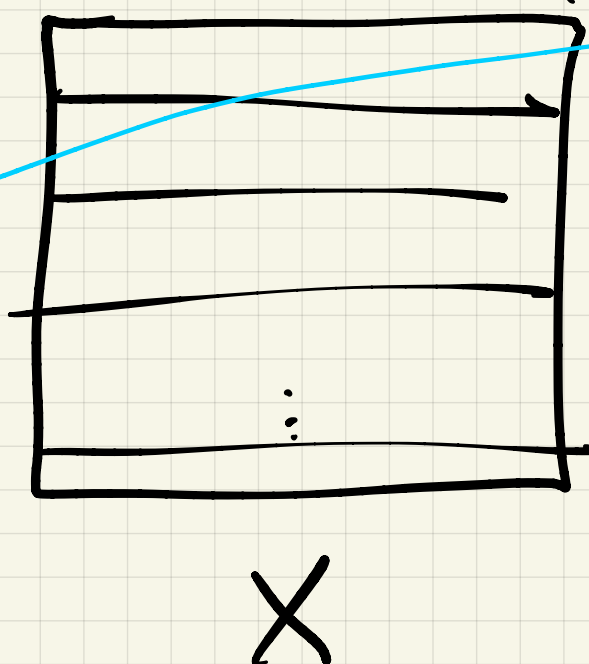
next token

d features

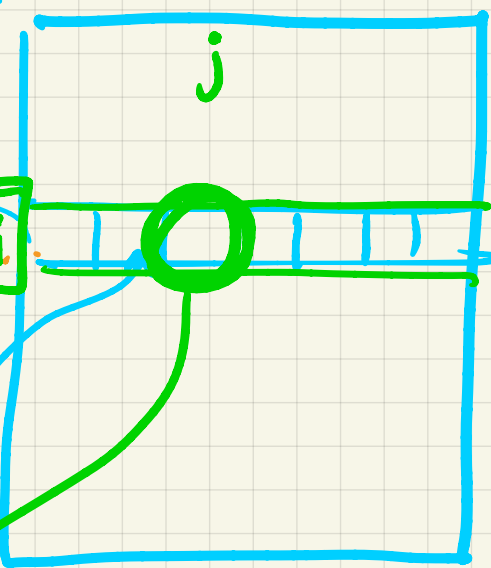
TL

TL

Once
upon
a
time
:
:
:
y



tokens in story



prob that token j
is the next token
after token i in the
story

$\hat{y}$

A transformer layer

